

Nested Classes vs Inheritance



"A nested class is also a member of its enclosing class" does member mean inherited class in java?



No, in Java, when it is said that "a nested class is also a member of its enclosing class," the term **member** does not mean that the nested class is **inherited** by the enclosing class. Instead, it means that the nested class is a part of (or belongs to) the enclosing class, much like fields (variables) and methods are members of a class.

Here's a breakdown to clarify:

Types of Nested Classes in Java:

1. Static Nested Classes

- A static nested class behaves like a static member of the enclosing class.
- It does not have access to non-static members (instance variables and methods) of the enclosing class.

2. Inner Classes (Non-static nested classes)

- An inner class behaves like an instance member of the enclosing class.
- It can access the enclosing class's members (fields and methods), including private ones.
- It is tied to an instance of the enclosing class and cannot be instantiated without an enclosing class instance.

Meaning of "Member":

- In the context of classes, a **member** is any element that belongs to the class, such as:
 - Fields (variables)
 - Methods
 - Constructors
 - Nested Classes (both static and inner classes)

Example:

```
java

public class Outer {
    private String outerField = "Outer";

    // Static Nested Class
    public static class StaticNested {
        public void display() {
            System.out.println("I am a static nested class.");
        }
    }

    // Inner Class
    public class Inner {
```

```

        public void display() {
            System.out.println("I am an inner class, and I can access: " + outerField);
        }
    }

    public static void main(String[] args) {
        // Accessing a static nested class
        Outer.StaticNested staticNested = new Outer.StaticNested();
        staticNested.display();

        // Accessing an inner class
        Outer outer = new Outer();
        Outer.Inner inner = outer.new Inner();
        inner.display();
    }
}

```

Inheritance in Java:

When a class is inherited, it involves extending a base (or parent) class using the `extends` keyword:

```

java

public class Parent {
    public void parentMethod() {
        System.out.println("I am a method in the parent class.");
    }
}

public class Child extends Parent {
    public void childMethod() {
        System.out.println("I am a method in the child class.");
    }
}

public class Test {
    public static void main(String[] args) {
        Child child = new Child();
        child.parentMethod(); // Inherited method
        child.childMethod(); // Child's own method
    }
}

```

Nested classes and inheritance are separate concepts in Java:

- Nested classes are about defining a class within another class for organizational or logical grouping.
- Inheritance is about creating a hierarchy where a child class reuses or overrides the behavior of a parent class.